

NAGRIK SETU AI

EVIDENCE-DRIVEN CIVIC INTELLIGENCE PLATFORM

From Citizen Problems to Evidence-Backed Civic Insights

A verifiable reasoning bridge between unstructured community grievances and open public datasets.

1	2	3	4	5	6
Input	NLP	Data	RAG	Synthesis	Decision
Citizen Problem	AI Parsing	Public Corpus	Verified Evidence	Grounded Analysis	Priority Signal

Stack: Multilingual NLP · Open Government Datasets · Hybrid Vector Retrieval · Deterministic Priority Scoring

Citizens Report Problems. But Context Is Missing.

DEFICIT 01

Unstructured Complaints

Natural language reports lack schema, standardized terminology, and spatial normalization.

DEFICIT 02

Fragmented Repositories

Infrastructure datasets sit isolated across departments, census releases, and municipal portals.

DEFICIT 03

Keyword Search Failures

Citizen dialect and informal descriptions fail to match formal civil engineering taxonomies.

DEFICIT 04

Isolated Decisions

Single grievances hide whether a localized failure reflects widespread ward-level distress.

Core Limitation: A complaint documents an individual lived experience, but cannot supply the public infrastructure evidence required to allocate municipal capital.

Introducing NagrikLens AI

An intelligence and evidence layer connecting citizen input to verifiable municipal action.

01	02	03	04	05
Understand	Discover	Verify	Analyze	Prioritize
Extracts entities, geo-bounds, category, and reported severity from raw text.	Indexes open-government datasets via multilingual semantic vector spaces.	Pins citations to specific dataset records, timestamps, and reporting agencies.	Constrains LLM reasoning strictly to retrieved factual public evidence.	Calculates transparent urgency through deterministic, rule-based scoring.

TRADITIONAL COMPLAINT MANAGEMENT

Citizen → Complaint Intake → Relational DB → Queue

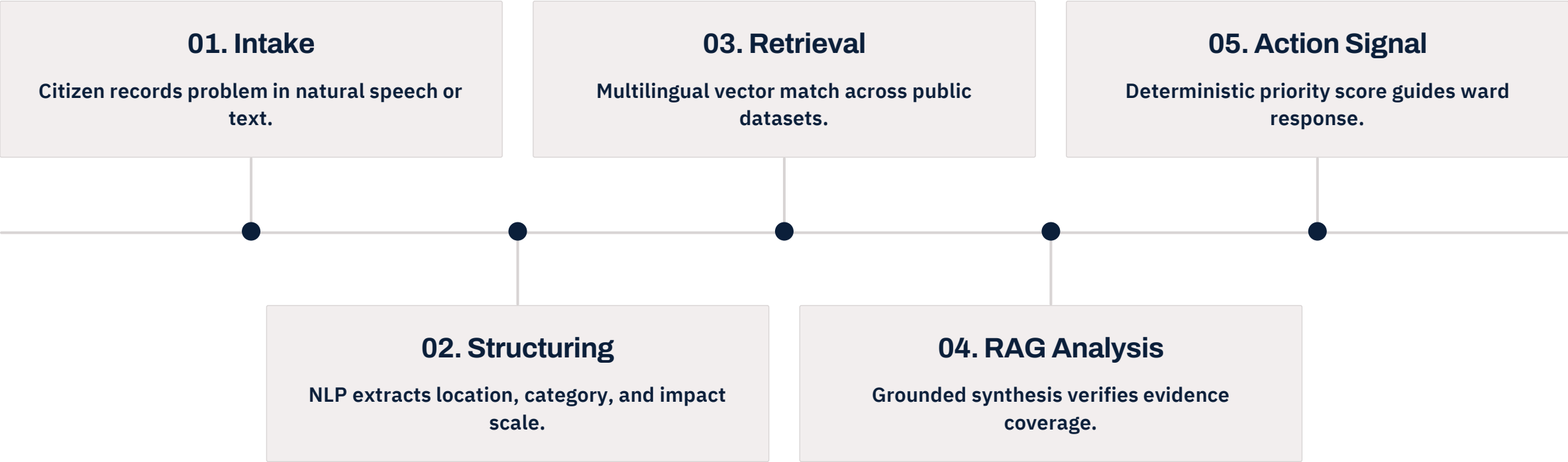
Static ticketing systems store text records without assessing infrastructure context.

NAGRIKLENS CIVIC INTELLIGENCE

Citizen → AI Parsing → Public Data → Evidence Grounding → Priority Signal

Transforms isolated complaints into contextualized, auditable civic intelligence.

From Citizen Voice to Civic Intelligence



SAMPLE REPORT "Many households in our village don't have reliable tap water"	STRUCTURED ENTITIES Domain: Water Infrastructure Geo: Rural Habitation Cohort: Multi-Household	PUBLIC MATCH Public Jal Jeevan Mission village scheme completion and pipeline records.	OUTPUT SIGNAL Documented pipeline gap identified. Priority: High. Evidence coverage: Strong.
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How NagrikLens Finds the Right Evidence

Pairing dense semantic embeddings with strict administrative metadata filtering.

PHASE 01

Entity Extraction

Engine: Gemini LLM

Parses unstructured text to isolate domain classification, administrative jurisdiction, reported severity, and affected population bounds.

PHASE 02

Semantic Index

Model: MiniLM-L12-v2

Generates 384-dimensional dense vectors via multilingual Sentence Transformers to bridge citizen vernacular and formal civil registries.

PHASE 03

Vector Search

Index: FAISS Vector Store

Executes sub-millisecond approximate nearest neighbor retrieval across chunked public infrastructure records and municipal registries.

PHASE 04

Metadata Filter

Filter: SQL Metadata

Enforces deterministic boundaries on state, district, department code, and dataset publication recency prior to RAG ingestion.

Retrieval Formula: Dense Vector Similarity + Administrative Metadata Filters → Precision Evidence Candidates

AI That Knows When the Evidence Is Not Enough

Citizen Request + Retrieved Evidence → RAG → Gemini → Evidence-Grounded Analysis

Constrained RAG

Gemini receives the citizen request and retrieved evidence, not free rein to invent information.

AI Output

Summary · Observations · Evidence Used · Limitations

Evidence Coverage

STRONG COVERAGE

PARTIAL COVERAGE

INSUFFICIENT COVERAGE

No evidence → No invented conclusion.

From Evidence to an Explainable Priority Signal

AI interpretation is separate from priority calculation.

Evidence → Factor Extraction → Deterministic Scoring → Priority Signal → Explanation

PROPOSED / CONFIGURABLE

Not an official government formula

Example factor weights

Severity	30%
Vulnerability	30%
Infrastructure Gap	25%
Evidence Coverage	15%

Principle: Missing data is represented as missing, never assumed.

Priority signal

High / Medium / Low

Supporting factors · Evidence used · Missing information · Explanation

TECHNICAL ARCHITECTURE

NagrikLens AI: System Architecture

Layer	Components
User layer	Citizen · Analyst · Admin
React frontend	React · Vite · TypeScript · Voice
FastAPI backend	FastAPI · Auth · Reports · Evidence · Analytics · Datasets
AI intelligence	Gemini · NLP · RAG
Retrieval engine	Sentence Transformers · FAISS · Metadata Search
Data layer	SQLite/PostgreSQL · Evidence · Reports · Datasets · Audit Logs
Verified public data	Open Government / Public Data

One Platform. Multiple Stakeholders.

Tailored operational surfaces driven by the same verified civic intelligence core.

FOR CITIZENS

Direct Civic Voice

Frictionless Reporting: Voice and text in English, Hindi, and Gujarati.

Contextual Feedback: Explanations of how their issue connects to local data.

Verifiable Tracking: Live status without administrative black boxes.

FOR ANALYSTS

Evidence Workbench

Semantic Exploration: Natural language search across complex datasets.

Grounded Synthesis: Pre-compiled citations linking complaints to records.

Deficit Analysis: Rapid detection of systemic ward infrastructure failures.

FOR ADMINISTRATORS

Decision Governance

Geographic Patterns: Cluster analysis identifying multi-ward bottlenecks.

Explainable Signals: Audit-ready priority rankings backed by public records.

Resource Allocation: Objective capital deployment defense for budgeting.

System Transformation: Moves civic operations from reactive complaint collection to evidence-aware infrastructure analysis.

The Web App Connects Reports to Public Evidence.

Evidence is surfaced as traceable records rather than as unsupported AI-generated claims.

NagrikLens AI

NagrikLens

CIVIC INTELLIGENCE

Dashboard

Report Problem

Evidence Explorer

Explore Issues

Priority Insights

Data Sources

Analytics

My Reports

Evidence Explorer

Search public evidence for: water access / Dharashiv / households

SEARCH

PUBLIC DATA

Water infrastructure record

Dataset • Geography • Reporting period

OPEN SOURCE

Household coverage evidence

Source organization • Dataset ID • URL

VERIFIED

Relevant supporting record

Verification status • Updated timestamp

PROV
ENAN
CE

Source
Dataset ID
Period
URL

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Semantic Retrieval + RAG = Evidence-Grounded Analysis.

The backend combines multilingual embeddings, vector search and metadata filtering before Gemini synthesizes the result.



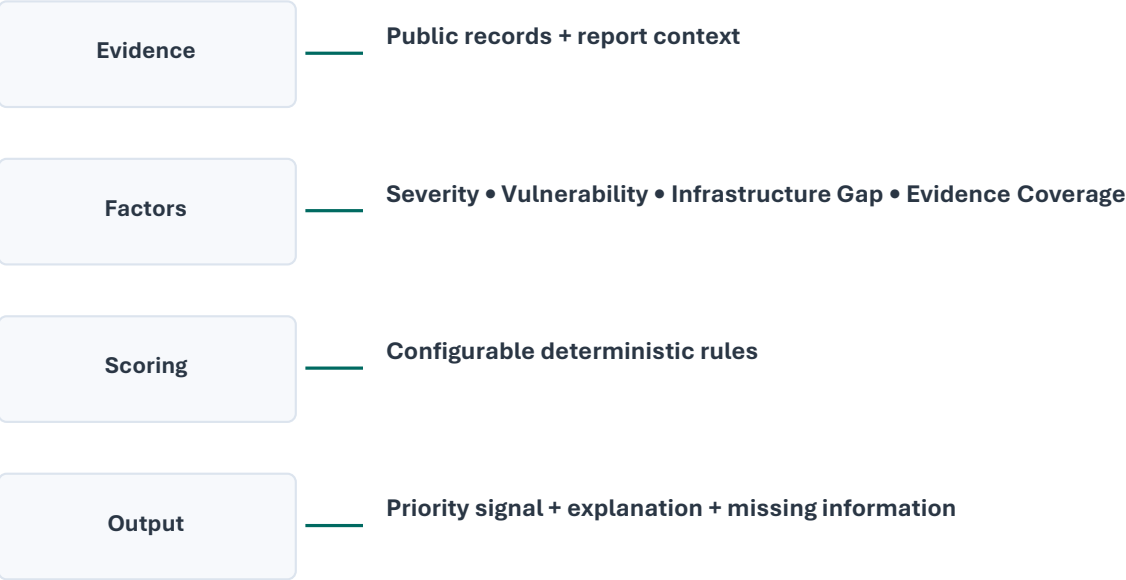
HYBRID RETRIEVAL
Semantic similarity + State/District/Category/Time metadata → relevant evidence candidates

No evidence → no invented conclusion.

AI Explains. Rules Calculate.

The web app separates evidence-grounded AI interpretation from the deterministic priority engine.

Priority Intelligence Flow



PROPOSED / CONFIGURABLE

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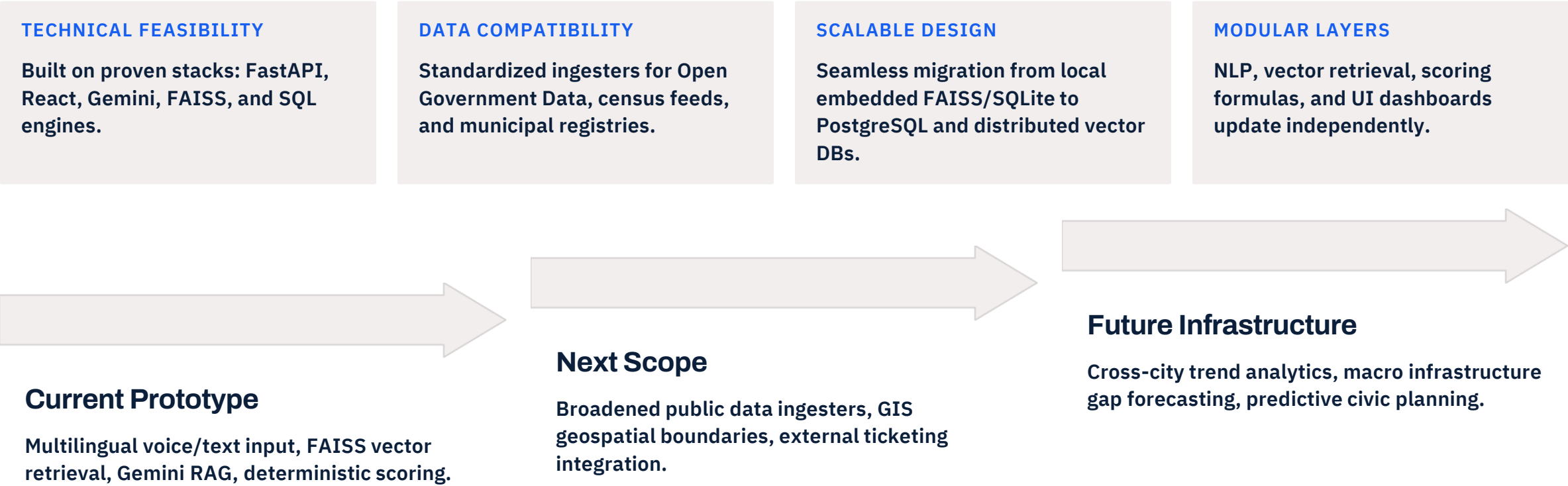
Missing information stays missing — it is never silently assumed.

Built as a Full-Stack AI Web Application.

The current frontend structure is designed to evolve into a backend-connected, dynamic platform.

FRONTEND	React • Vite • TypeScript • Responsive UI • Voice input
BACKEND	Python • FastAPI • Authentication • Reports • Evidence • Analytics
AI	Google Gemini • Structured extraction • RAG analysis
RETRIEVAL	Sentence Transformers • FAISS • Hybrid metadata search
DATA	SQLite / PostgreSQL-compatible design • Reports • Evidence • Datasets • Audit logs
PUBLIC DATA	Open government / verified public datasets with provenance

From Prototype to Civic Infrastructure



See the Problem. Find the Evidence. Understand the Context.

Citizen Voice + Public Data + AI = Evidence-Driven Civic Intelligence

**Thank
You**